

APPENDIX C: PATH SAMPLING ALGORITHM AND ANALYSIS

Here we present an algorithm for path sampling and provide a proof that it leads to an unbiased estimate for the gradient. The main idea is to sample edges from the set of outgoing edges for each vertex in topological order, and scale appropriately. Vertices that have no incoming edges sampled can be skipped.

Algorithm 1 RMAD with path sampling

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1: Inputs:
2:    $\mathcal{G} = (V, E)$  - Computational Graph.  $d_v$  denotes outdegree,  $v.succ$  successor set of vertex  $v$ .
3:    $y$  - Output vertex
4:    $\Theta = (\theta_1, \theta_2, \dots, \theta_m) \subset V$  - Input vertices
5:    $k > 0$  - Number of samples per vertex
6: Initialization:
7:    $Q(e) = 0, \forall e \in E$ 
8: for  $v$  in topological order; synchronous with forward computation do
9:   if No incoming edge of  $v$  has been sampled then
10:    Continue
11:   for  $k$  times do
12:    Sample  $i$  from  $[d_v]$  uniformly.
13:     $Q(v, v.succ[i]) \leftarrow Q(v, v.succ[i]) + \frac{d_v}{k} \frac{\partial v.succ[i]}{\partial v}$ 
14: Run backpropagation from  $y$  to  $\Theta$  using  $Q$  as intermediate partials.
15: Output:  $\nabla_{\Theta} y$ 

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The main storage savings from Algorithm 1 will come from Line 9, where we only consider a vertex if it has an incoming edge that has been sampled. In computational graphs with a large number of independent paths, this will significantly reduce memory required, whether we record intermediate variables and recompute the LCG, or store entries of the LCG directly.

To see that path sampling gives an unbiased estimate, we use induction on the vertices in reverse topological order. For every vertex z , we denote $\bar{z} = \frac{\partial y}{\partial z}$ and $\hat{z}$ as our approximation for $\bar{z}$. For our base case, we let $\hat{y} = \frac{dy}{dy} = 1$, so $\mathbb{E}\hat{y} = \bar{y}$. For all other vertices z , we define

$$\hat{z} = d_z \sum_{(z,v) \in E} \mathbb{I}_{v=v_i} \frac{\partial v}{\partial z} \hat{v} \quad (9)$$

where d_z is the out-degree of z , v_i is sampled uniformly from the set of successors of z , and $\mathbb{I}_{v=v_i}$ is an indicator random variable denoting if $v = v_i$. We then have

$$\mathbb{E}\hat{z} = \sum_{(z,v) \in E} d_z \mathbb{E}[\mathbb{I}_{v=v_i}] \frac{\partial v}{\partial z} \mathbb{E}\hat{v} = \sum_{(z,v) \in E} \frac{\partial v}{\partial z} \mathbb{E}\hat{v} \quad (10)$$

assuming that the randomness over the sampling of outgoing edges is independent of $\hat{v}$, which must be true because our induction is in reverse topological order. Since by induction we assumed $\mathbb{E}\hat{v} = \bar{v}$, we have

$$\mathbb{E}\hat{z} = \sum_{(z,v) \in E} \frac{\partial v}{\partial z} \bar{v} = \bar{z} \quad (11)$$

which completes the proof.